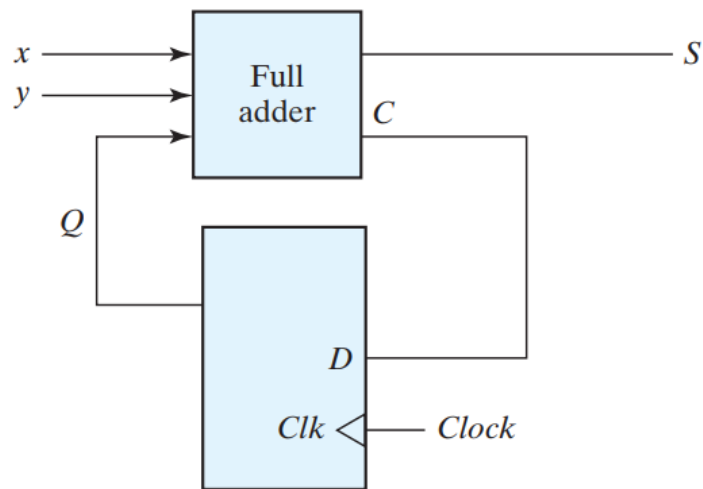


Digital systems

Tutorial 6

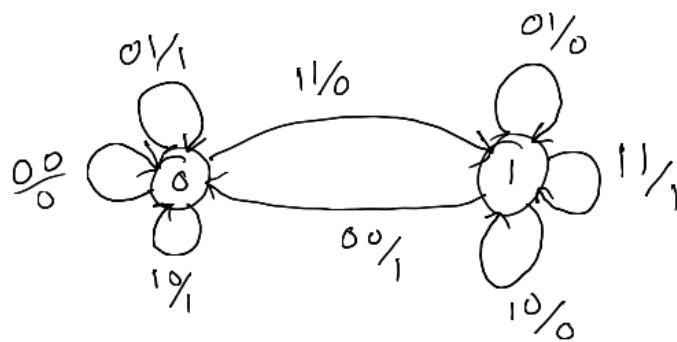
March 2023

1. A sequential circuit has one flip-flop Q , two inputs x and y , and one output S . It consists of a full-adder circuit connected to a D flip-flop, as shown in the figure below. Derive the state table and state diagram of the sequential circuit.

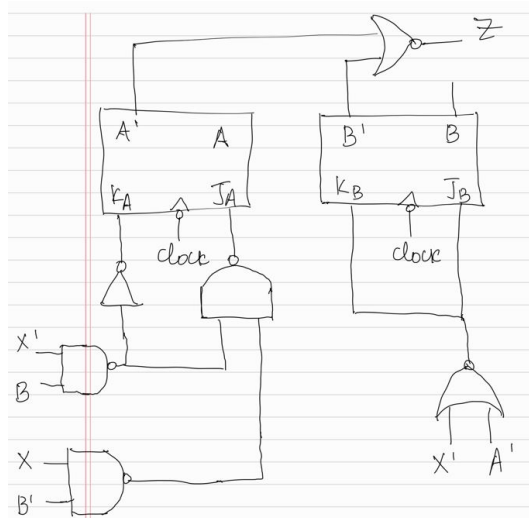


1. state table

Q^t	x	y	Q^{t+1}	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1



2. (a) For the following sequential circuit find the next state equation. Is this a Mealy or Moore machine? Using these next state equations, construct a state table and graph for the circuit.
- (b) What is the output sequence when the input sequence is $X = 01100$?
- (c) Draw the timing diagram for the input sequence in (b). Show the clock, X, A, B and Z . Assume that the input changes between falling and rising clock edges.



Solution:- Characteristic eqn of J-K FF:-

$$Q_{n+1} = J\bar{Q} + KQ$$

$$J_A = X\bar{B} + \bar{X}B$$

$$Q_A^+ = A(\bar{B} + X) + \bar{A}(B\bar{X} + \bar{B}X)$$

$$K_A = \bar{X}B$$

$$Q_B^+ = A\bar{B}X + B(\bar{A} + \bar{X})$$

$$J_B = K_B = X \cdot A$$

$$Z = A \cdot B$$

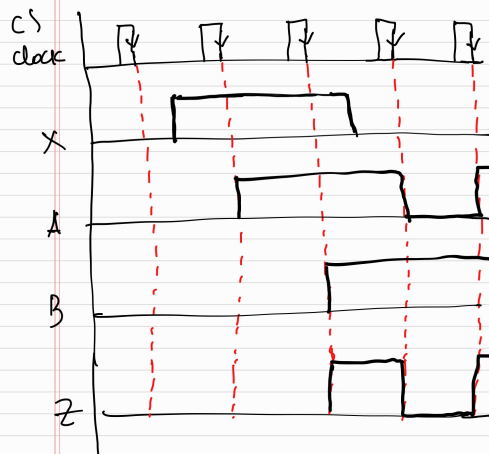
⇒ Moore machine

Input	Present state		Next state		O/p
X	A	B	A ⁺	B ⁺	Z
0	0	0	0	0	0
0	0	1	1	1	0
0	1	0	1	0	0
0	1	1	0	1	1
1	0	0	1	0	0
1	0	1	0	1	0
1	1	0	1	1	0
1	1	1	1	0	1

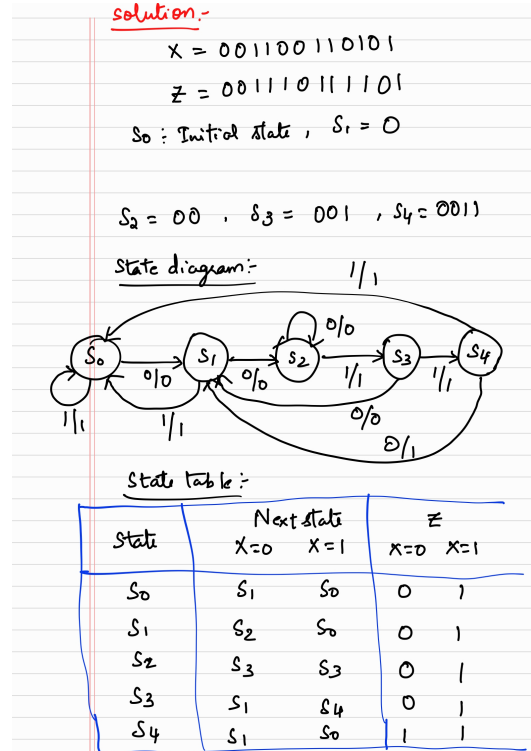
(b) Input sequence = 01100

Z = 00101

J	K	Q _{n+1}
0	0	Q _n
0	1	0
1	0	1
1	1	Q _{n+1}



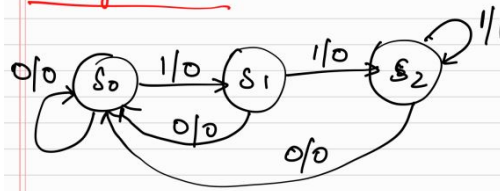
3. A Mealy sequential circuit has one input (X), one output (Z). The circuit should transmit its input, except that it should prevent the sequence 00110 from occurring. So, Z should be the same as X, except that if the input sequence 00110 occurs, Z should be 1 rather than 0 when the last 0 is received, so that the sequence $X=00110$ is replaced with $Z=00111$. Derive the state graph and table for input sequence $X = 001100110101$



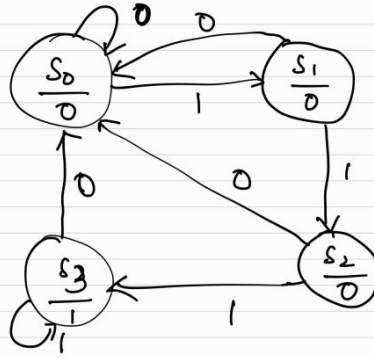
4. Implement a sequence detector circuit to detect the sequence detector '111' using Mealy and Moore

Solution:-

Mealy Model:-



Moore Model:-



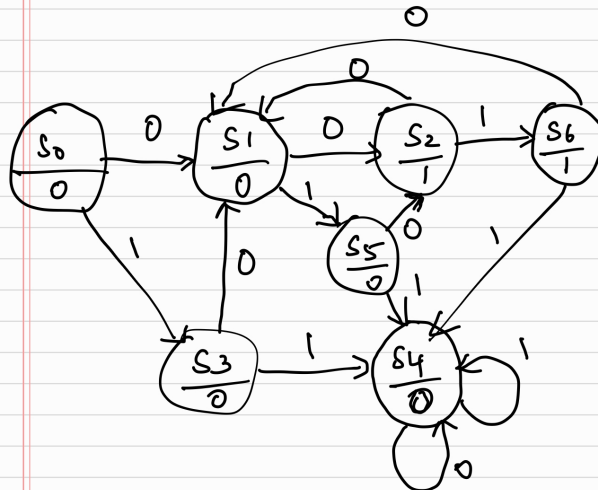
5. A clocked Moore sequential circuit has an output of $Z = 1$ if the total number of 0's received is an even number greater than zero, provided that two consecutive 1's have never been received.

Solution:-

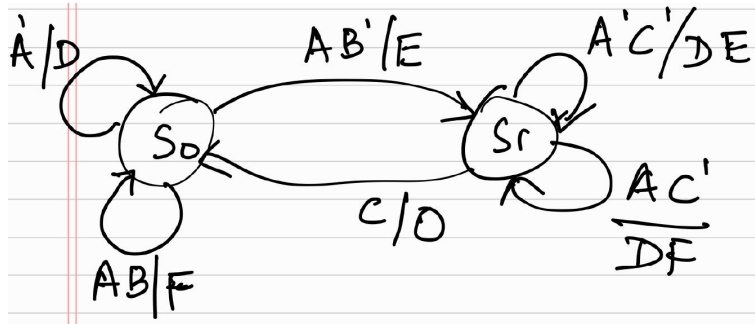
To understand the problem statement, the O/P sequence for the following input sequence.

$X = 00001010110000$

$Z = (0)01011001100000$



6. For the following state graph, construct the state table.



Solution:-

		Next State							
		000	001	010	011	100	101	110	111
S ₀		S ₀	S ₀	S ₀	S ₀	S ₁	S ₁	S ₀	S ₀
S ₁		S ₁	S ₀	S ₁	S ₀	S ₁	S ₀	S ₁	S ₀
		output (DEF)							
		000	001	010	011	100	101	110	111
S ₀		100	100	100	100	010	010	001	001
S ₁		110	000	110	000	101	000	101	000

7. A new flip-flop has the following truth table where X and Y are the inputs

X	Y	Q_{n+1}
0	0	0
0	1	$\overline{Q_n}$
1	0	Q_n
1	1	1

Determine the following

- The characteristic table
- The characteristic equation
- The excitation table

12.

(a) Characteristic table

Q_n	X	Y	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

(b)

$Q_n \backslash XY$	00	01	11	10
0	0	1	0	0
1	0	0	1	1

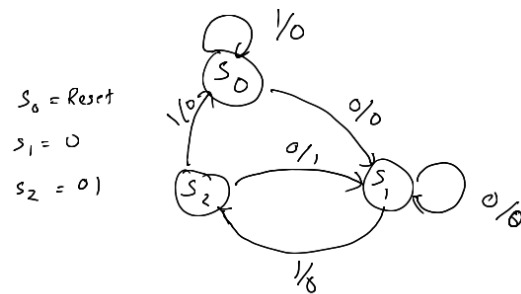
$$Q_{n+1} = \overline{Q_n} Y + Q_n X$$

(c) excitation table

Q_n	Q_{n+1}	X	Y
0	0	x	0
0	1	x	1
1	0	0	x
1	1	1	x

8. Design a sequence detector which outputs high when the pattern **010** is detected from an input bit stream.

(14) state diagram



State assignment

$$S_0 = 00$$

$$S_1 = 01$$

$$S_2 = 10$$

State table

q_1	q_2	u	q_1'	q_2'	y
0	0	0	0	1	0
0	0	1	0	0	0
0	1	0	0	1	0
0	1	1	1	0	0
1	0	0	0	1	1
1	0	1	0	0	0

Using D-flip flop

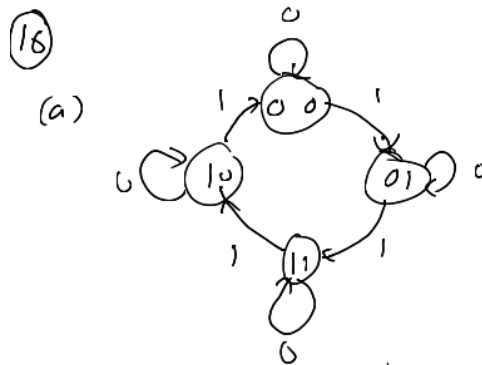
$$D_1 = \bar{q}_1 q_2 u$$

$$D_2 = q_1 \bar{q}_2 u = u'$$

$$y = q_1 \bar{q}_1 \bar{u}$$

9. Design a sequential circuit with two D flip-flops A and B , and one input x_{in} .

- (a) When $x_{in} = 0$ the state of the circuit remains the same. When $x_{in} = 1$, the circuit goes through the state transitions from $00 \rightarrow 01 \rightarrow 11 \rightarrow 10 \rightarrow 00$ and repeat.
- (b) When $x_{in} = 0$ the state of the circuit remains the same. When $x_{in} = 1$, the circuit goes through the state transitions from $00 \rightarrow 11 \rightarrow 01 \rightarrow 10 \rightarrow 00$ and repeat.

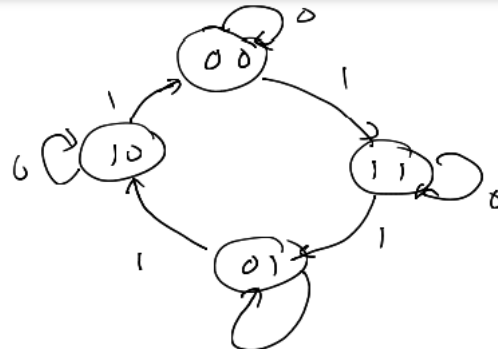


Q_1	Q_2	x_{in}	Q_1'	Q_2'
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	1
1	0	0	1	0
1	0	1	0	0
1	1	0	1	1
1	1	1	1	0

$$D_1 = \begin{array}{c|ccc} & x_{in}=0 & x_{in}=1 & \\ \hline Q_2 & 0 & 1 & \\ \hline 0 & 0 & 0 & \\ 1 & 0 & 1 & \end{array} = Q_2 x_{in} + Q_1 \overline{x_{in}}$$

$$D_2 = \begin{array}{c|ccc} & x_{in}=0 & x_{in}=1 & \\ \hline Q_1 & 0 & 1 & \\ \hline 0 & 0 & 1 & \\ 1 & 0 & 0 & \end{array} = \overline{Q_1} x_{in} + Q_2 \overline{x_{in}}$$

(b)



Q_1	Q_2	n_{in}	Q_1'	Q_2'
0	0	0	0	0
0	0	1	1	1
0	1	0	0	1
0	1	1	1	0
1	0	0	1	0
1	0	1	0	0
1	1	0	1	1
1	1	1	0	1

$$D_1 = \begin{matrix} & Q_2 n_{in} \\ Q_1 & \begin{array}{cc|cc} 00 & 01 & 11 & 10 \\ \hline 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{array} \end{matrix} = Q_1 \bar{n}_{in} + \bar{Q}_1 n_{in}$$

$$D_2 = \begin{matrix} & Q_1 n_{in} \\ Q_2 & \begin{array}{cc|cc} 00 & 01 & 11 & 10 \\ \hline 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{array} \end{matrix} = Q_2 \bar{n}_{in} + n_{in} (Q_1 \oplus Q_2)$$

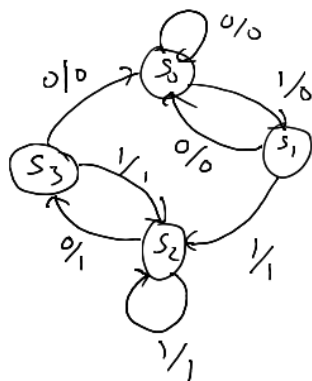
10. It is required to eliminate short length pulses from sampled data. It means 0's in continuous 1's have to be made 1 & similarly 1's in continuous 0's are to be made 0's as shown in the following example. Draw the state machine and derive the state equation for the same.

Eg:

I/P 0 1 0 0 1 1 0 1 1 0 0

O/P 0 0 0 0 0 1 1 1 1 1 0

20.



State assignment

$S_0 = 00$, $S_1 = 01$, $S_2 = 10$, $S_3 = 11$

Q_1	Q_2	u	Q_1'	Q_2'	y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	0
0	1	1	1	0	1
1	0	0	1	1	0
1	0	1	1	0	1
1	1	0	0	0	0
1	1	1	1	0	1

Q_1, Q_2

00	01	11	10
0	0	1	0
0	1	0	0
1	0	1	1

$$Q_1' = Q_1 \bar{Q}_2 + Q_2 u$$

Q_2'

0	1	0	0
0	0	0	0
1	0	0	0
1	1	0	0

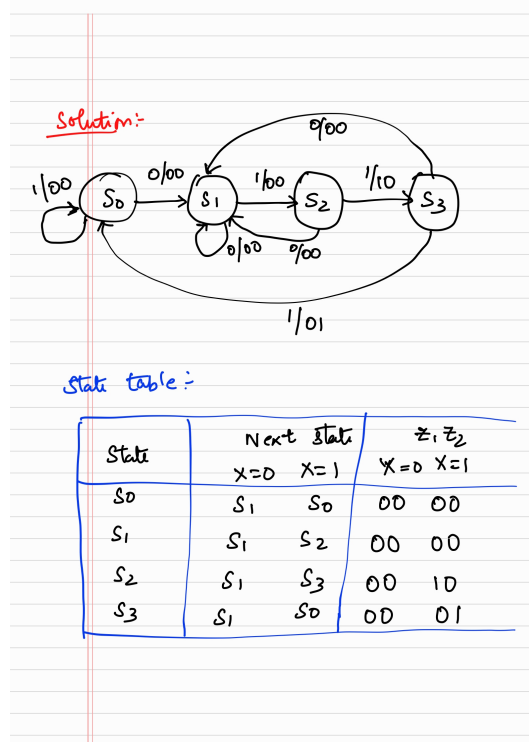
$$Q_2' = \bar{Q}_2 (Q_1 \oplus u)$$

y

0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0

$$y = Q_1 u + Q_2 u$$

11. A Mealy sequential circuit has one input (X) and two outputs (Z1 and Z2). The circuit produces an output of Z1 = 1 whenever sequence 011 is completed and an output of Z2 = 1 whenever sequence 0111 is completed. Derive the state graph and table.



12. A sequential circuit has the form of $D1 = Q2\overline{Q3}$, $D1 = Q3$, $D3 = \overline{Q2} + X$, $Z = X\overline{Q2} + \overline{X}Q2$
- Is this a Mealy or Moore machine?
 - Construct state table and state graph for the circuit
 - Draw a timing diagram for the circuit showing the clock, X, Q1, Q2, Q3, and Z. Use the input sequence X = 01011. Change X between clock edges.

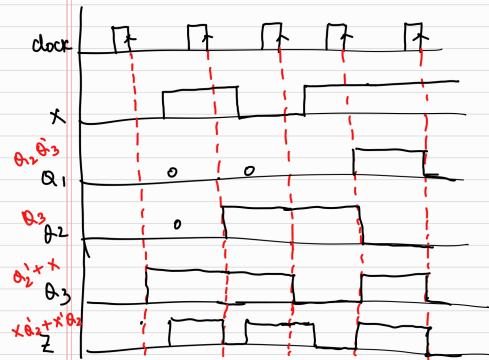
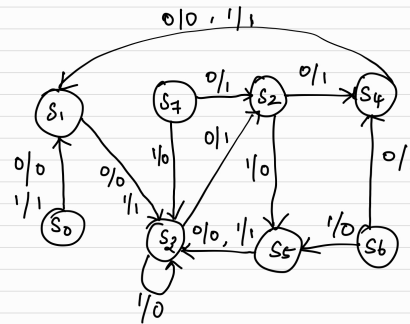
Solution:-

(a) In Moore, output doesnot depend on input. $\therefore$ This is a Mealy Machine.

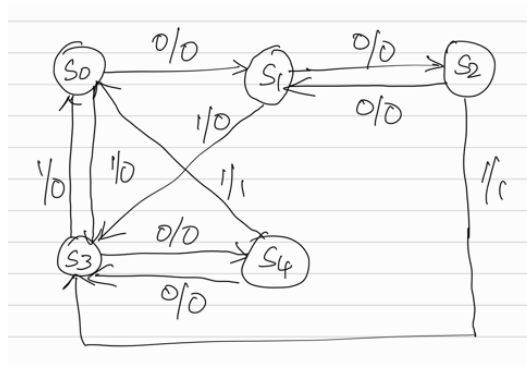
$$Q_{n+1} = D$$

(b)

	Q_1, Q_2, Q_3	Q_1^+, Q_2^+, Q_3^+			Z	
		Q_1^+ $X=0$	Q_2^+ $X=1$	Q_3^+ $X=1$	$X=0$	$X=1$
S_0	0 0 0	0 0 1	0 0 1	0 0 1	0	1
S_1	0 0 1	0 1 1	0 1 1	0 1 1	0	1
S_2	0 1 0	1 0 0	1 0 1	1 0 1	1	0
S_3	0 1 1	0 1 0	0 1 1	0 1 1	1	0
S_4	1 0 0	0 0 1	0 0 1	0 0 1	0	1
S_5	1 0 1	0 1 1	0 1 1	0 1 1	0	1
S_6	1 1 0	1 0 0	1 0 1	1 0 1	1	0
S_7	1 1 1	0 1 0	0 1 1	0 1 1	1	0



13. For the following state graph, give the state table. Then, give the timing diagram for the input sequence $X = 101001$. Assume X changes midway between the falling and rising edge of the clock, and that the flip flops are falling edge triggered. What is the correct output sequence?

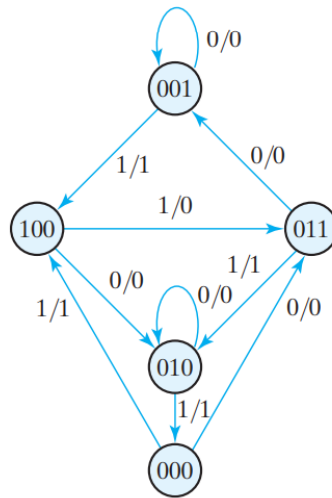


Solution:-

State	Present state $a_1 a_2 a_3$	Next state $a_1^+ a_2^+ a_3^+$		Z	
		$X=0$	$X=1$	$X=0$	$X=1$
S_0	0 0 0	0 0 1	0 1 1	0	0
S_1	0 0 1	0 1 0	0 1 1	0	0
S_2	0 1 0	0 0 1	0 1 1	0	1
S_3	0 1 1	1 0 0	0 0 0	0	0
S_4	1 0 0	0 1 1	0 0 0	0	1



14. A sequential circuit has three flip-flops A, B, C ; one input x_{in} and one output y_{out} . The state diagram is as shown below. The circuit is to be designed by treating the unused states as don't-care conditions using T flip-flops.



(17)

A	B	C	x_{in}	A'B'C'	y	T_A	T_B	T_C
0	0	0	0	011	0	0	1	1
0	0	0	1	100	1	1	0	0
0	0	1	0	001	0	0	0	0
0	0	1	1	100	1	1	0	1
0	1	0	0	010	0	0	0	0
0	1	0	1	000	1	0	1	0
0	1	1	0	001	0	0	1	0
0	1	1	1	010	1	0	0	1
1	0	0	0	010	0	1	1	0
1	0	0	1	011	0	1	1	1

T_A x_{in}

00	01	11	10
00	0	0	0
01	0	0	0
11	0	0	0
10	1	1	0

$T_A = A + \bar{B}x_{in}$

T_B

00	01	11	10
00	0	0	0
01	0	0	0
11	0	0	0
10	1	1	0

$T_B = A + \bar{B}\bar{C}x_{in} + B\bar{C}x_{in} + B\bar{C}x_{in}$

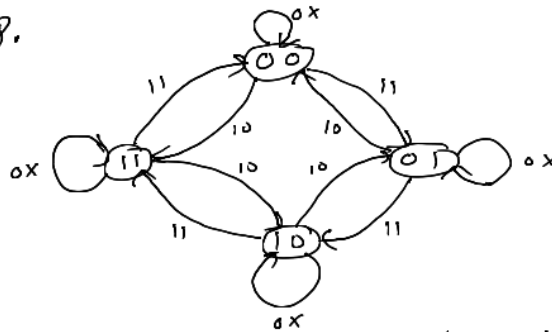
T_C

00	01	11	10
00	0	0	0
01	0	0	0
11	0	0	0
10	1	1	0

$T_C = Ax_{in} + x_{in}C + \bar{A}\bar{B}\bar{C}x_{in}$

15. Design a sequential circuit with two JK flip-flops A and B and two inputs E and F . If $E = 0$, the circuit remains in the same state regardless of the value of F . When $E = 1$ and $F = 1$, the circuit goes through the state transitions from 00 to 01, to 10, to 11, back to 00, and repeats. When $E = 1$ and $F = 0$, the circuit goes through the state transitions from 00 to 11, to 10, to 01, back to 00, and repeats.

18.



A	B	E	F	A'	B'	J_A	K_A	J_B	K_B
0	0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0	0	0
0	0	1	0	1	1	1	0	1	0
0	0	1	1	0	1	0	0	1	0
0	1	0	0	0	1	0	0	0	0
0	1	0	1	0	0	0	0	0	0
0	1	1	0	1	0	1	0	1	0
0	1	1	1	1	0	1	0	1	0
1	0	0	0	1	0	0	0	0	0
1	0	0	1	0	1	0	0	0	0
1	0	1	0	1	1	0	0	0	0
1	0	1	1	1	1	0	0	0	0
1	1	0	0	1	1	0	0	0	0
1	1	0	1	1	0	0	0	0	0
1	1	1	0	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0

AB	EF	00	01	11	10
00	00	0	0	0	0
01	01	0	0	0	0
11	11	0	0	0	0
10	10	0	0	0	0

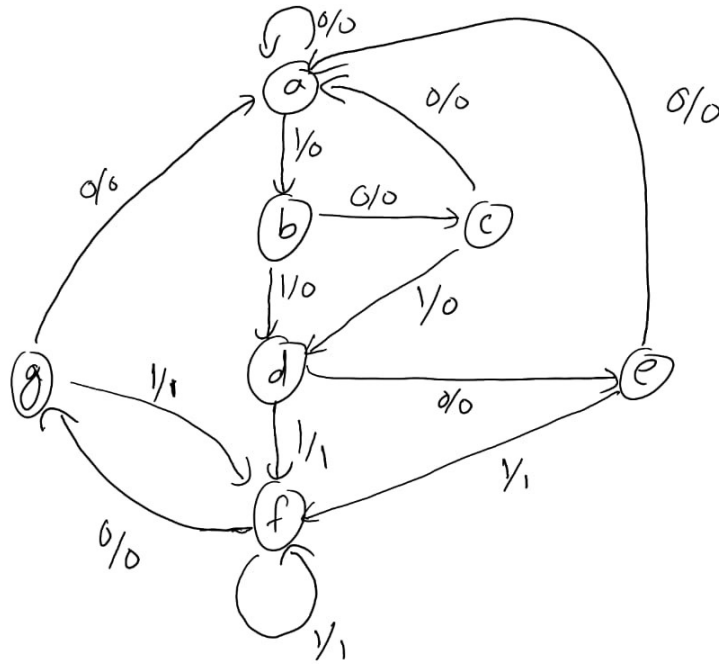
AB	EF	00	01	11	10
00	00	0	0	0	0
01	01	0	0	0	0
11	11	0	0	0	0
10	10	0	0	0	0

AB	EF	00	01	11	10
00	00	0	0	0	0
01	01	0	0	0	0
11	11	0	0	0	0
10	10	0	0	0	0

AB	EF	00	01	11	10
00	00	0	0	0	0
01	01	0	0	0	0
11	11	0	0	0	0
10	10	0	0	0	0

$J_A = K_A = BEF + \bar{B}\bar{E}\bar{F}$ and $J_B = K_B = E$

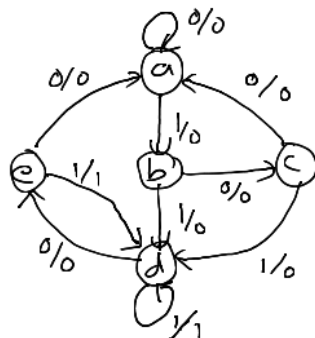
16. For the given state diagram do the following



- Identify the type of state machine and write the state table.
- Perform state reduction and draw the reduced state diagram.
- Implement the reduced state machines using T flip-flops.

19. The given machine is a mealey machine
state table

P.S	N.S		Output	
	n=0	n=1	n=0	n=1
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	a d	0	1
e	a	a d	0	1
f	e g f		0	1
g	a	f	0	1



Reduced
state machine

state assignment

a = 000

b = 001

c = 010

d = 011

e = 100

∴ 3 flip flops required.

Q_A	Q_B	Q_C	u	$Q_A'Q_B'Q_C'$	y	T_A	T_B	T_C
0	0	0	0	000	0	0	0	0
0	0	0	1	001	0	0	0	1
0	0	1	0	010	0	0	1	1
0	0	1	1	011	0	0	1	0
0	1	0	0	000	0	0	1	0
0	1	0	1	011	0	0	0	1
0	1	1	0	100	0	1	1	1
0	1	1	1	011	1	0	0	0
1	0	0	0	000	0	1	0	0
1	0	0	1	011	1	1	1	1

$$y = \bar{Q}_A Q_B Q_C u + Q_A \bar{Q}_B \bar{Q}_C u$$

T_A

$Q_C u$	00	01	11	10
$Q_A Q_B$				
00				
01				
11	0	0	0	0
10	1	1	1	1

$$T_A = Q_A + Q_B Q_C \bar{u}$$

T_B

$Q_C u$	00	01	11	10
$Q_A Q_B$				
00				
01				
11	0	0	0	0
10	1	1	1	1

$$T_B = Q_B \bar{u} + Q_A u + Q_C \bar{u} + Q_B Q_C$$

T_C

$Q_C u$	00	01	11	10
$Q_A Q_B$				
00				
01				
11	0	0	0	0
10	1	1	1	1

$$T_C = Q_C u + Q_C \bar{u} = Q_C$$

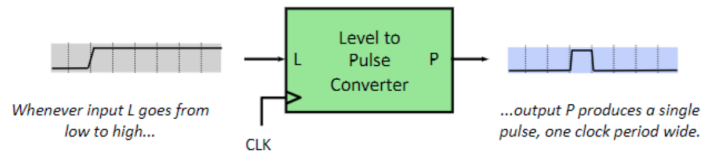
y

$Q_C u$	00	01	11	10
$Q_A Q_B$				
00				
01				
11	0	0	0	0
10	1	1	1	1

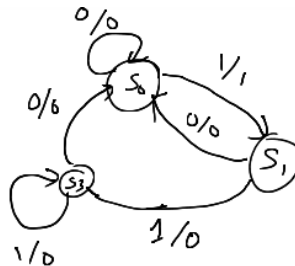
$$y = Q_A u + Q_B Q_C u$$

17. A level to pulse converter is to be designed using sequential circuits.

- Draw the state diagram
- Write the state equation



13.



State assignment².

$$S_0 = 00, S_1 = 01, S_2 = 10$$

q_1	q_2	u	q_1'	q_2'	y
0	0	0	0	0	0
0	0	1	0	1	1
0	1	0	0	0	0
0	1	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	0

$$q_1'$$

q_2	u	q_1'
00	01	10
01	00	01
01	01	10
10	00	01
10	01	10

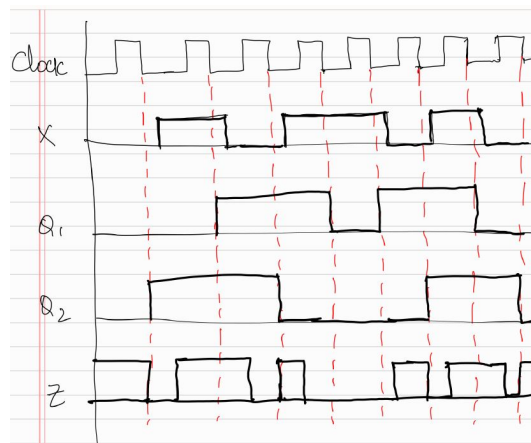
$$q_1' = q_1 u + q_2 u$$

$$q_2'$$

q_1	u	q_2'
00	01	10
01	00	01
01	01	10
10	00	01
10	01	10

$$q_2' = y = \bar{q}_1 \bar{q}_2 u$$

18. A Mealy sequential circuit has one input, one output and two flip flops. A timing diagram for the circuit follows. Construct a state table and state graph for the circuit?



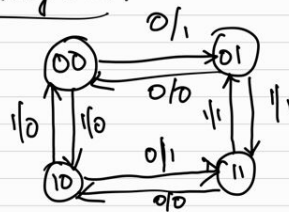
Solution:-

clock cycle	Information Gathered
1	$Q_1, Q_2 = 00, X=0 \Rightarrow Z=1, Q_1^+, Q_2^+ = 01$
2	$Q_1, Q_2 = 01, X=0 \Rightarrow Z=0; X=1 \Rightarrow Z=1, Q_1^+, Q_2^+ = 11$
3	$Q_1, Q_2 = 11, X=1 \Rightarrow Z=1; X=0 \Rightarrow Z=0, Q_1^+, Q_2^+ = 10$
4	$Q_1, Q_2 = 10, X=0 \Rightarrow Z=1; X=1 \Rightarrow Z=0, Q_1^+, Q_2^+ = 00$
5	$Q_1, Q_2 = 00, X=1 \Rightarrow Z=0, Q_1^+, Q_2^+ = 10$
6	$Q_1, Q_2 = 10, X=1 \Rightarrow Z=0; X=0 \Rightarrow Z=1, Q_1^+, Q_2^+ = 11$
7	$Q_1, Q_2 = 11, X=0 \Rightarrow Z=0; X=1 \Rightarrow Z=1, Q_1^+, Q_2^+ = 01$
8	$Q_1, Q_2 = 01, X=1 \Rightarrow Z=1; X=0 \Rightarrow Z=0, Q_1^+, Q_2^+ = 00$
9	$Q_1, Q_2 = 00, X=0 \Rightarrow Z=1$

State table:-

Present state Q_1, Q_2	Next state Q_1^+, Q_2^+		Z
	X=0	X=1	
0 0	01	10	1 0
0 1	00	11	0 1
1 0	11	00	1 0
1 1	10	01	0 1

state diagram:-



19. Reduce the following state table to a minimum number of states

Present state	Next state		Present o/p	
	X=0	X=1	X=0	X=1
a	h	c	1	0
b	c	d	0	1
c	h	b	0	0
d	f	h	0	0
e	c	f	0	1
f	f	g	0	0
g	g	c	1	0
h	a	c	1	0

Solution:-

b							
c							
d							
e							
f							
g							
h							
a							

From the table, $a \equiv g \equiv h$,
 $d \equiv f$, $e \equiv b$

Reduced state table:-

State	Next state		Output	
	X=0	X=1	X=0	X=1
a	a	c	1	0
b	c	d	0	1
c	a	b	0	0
d	d	a	0	0

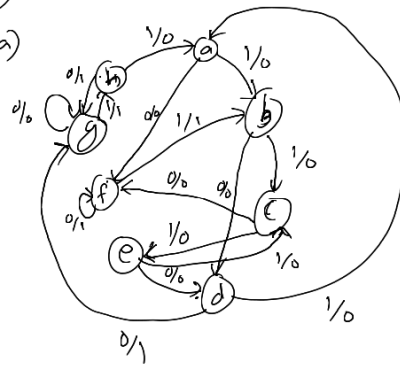
20. For the following state table

Present State	Next State		Output	
	$x = 0$	$x = 1$	$x = 0$	$x = 1$
<i>a</i>	<i>f</i>	<i>b</i>	0	0
<i>b</i>	<i>d</i>	<i>c</i>	0	0
<i>c</i>	<i>f</i>	<i>e</i>	0	0
<i>d</i>	<i>g</i>	<i>a</i>	1	0
<i>e</i>	<i>d</i>	<i>c</i>	0	0
<i>f</i>	<i>f</i>	<i>b</i>	1	1
<i>g</i>	<i>g</i>	<i>h</i>	0	1
<i>h</i>	<i>g</i>	<i>a</i>	1	0

- (a) Draw the corresponding state diagram.
- (b) Tabulate the reduced state table.
- (c) Draw the state diagram corresponding to the reduced state table.

15:

(a)



(b)

Present state	Next state		O/P	
	n=0	n=1	n=0	n=1
a	f	b	0	0
b	d	a	0	0
c	f	b	0	0
d	g	a	1	0
e	d	c	0	0
f	f	b	1	1
g	g	a	1	0

(c) Reduced state diagrams

